

Hao Pan

AI Full-Stack Engineer | Agent Applications & Data Systems

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Summary

Master of Computing candidate in Applied AI at NTU, focused on AI full-stack and Agent applications: turning real business problems into AI applications that run reliably. I build the frontend, backend, and data pipeline with Python and TypeScript, and keep failure-critical parts reliable with SQL templates, structured outputs, and explicit failure handling.

Education

Nanyang Technological University

Nov 2025 - Jan 2027 (Expected)

Master of Computing in Applied Artificial Intelligence, College of Computing and Data Science

Singapore

Suzhou University

Sep 2019 - Jun 2023

B.Eng. in Data Science and Big Data Technology; National Scholarship (top 0.2% nationwide)

China

Experience

Shopee Singapore

Mar 2026 - Present (through Sep 2026)

AI Data Analyst Intern, Buyer & Marketing Business Intelligence

Singapore

Deliver AI applications for the Buyer & Marketing BI team, targeting an end-to-end flow from PRD to data request to data Agent to result. The bottleneck isn't query accuracy – it's that PMs without a SQL background can't write an accurate request or find an existing, mature Topic. I've delivered and shipped this link: unifying historical requests, business SQL, and Diana Topics into one knowledge base, so PMs produce a structured, traceable request in natural language and route it to the correct Topic.

Analysis Agent Copilot Full-stack AI application for analysts; in daily use

- **Full-Stack Delivery:** Owned the full cycle – requirements, architecture, delivery, and iteration – across a Svelte frontend, a layered FastAPI backend (routes / services / repositories / renderers), a Python ingestion pipeline, and Git-versioned storage.
- **Agent Architecture:** The LLM is scoped to intent understanding and candidate explanation; the Router calls one Handler per turn from a static allowlist (Tool Use). The retrieve → select → authorize → generate state machine is fully persisted, evidence bound to a Git commit and content hash, with a Changed / Preserved / Missing diff shown before generation.
- **Delivery & Evaluation:** In daily use by analysts; curated 69 source- and version-bound historical cases forming the metric definitions and SQL foundation behind every generated request. Designed a retrieval evaluation on Recall@5, nDCG@5, and citation accuracy; results drove adoption of the open-source Agent framework Pi (pi.dev) as the orchestration layer, now completing performance tuning before rollout.

A/B Test Agent Skills Internal Agent platform; used in business analysis

- **Skill Engineering & Acceptance:** Co-designed and executed an SOP for building and validating Data Agent topics, covering table selection, knowledge-base curation, a 40+ question UAT test set, and error-taxonomy-driven regression fixes, meeting the accuracy target; packaged recurring A/B test analysis across multi-country BI teams into reusable Agent Skills, used in real business analysis.

Jinken Vocational and Technical College

Mar 2023 - Oct 2025

Lecturer & AI Technology Lead

China

- Taught general Python and AI courses to 300+ students; coached the team to Third Prize at the 2024 RAICOM National Finals.

Selected Projects

ContextCue (Agent Memory Layer Prototype)

Jun 2026 - Jul 2026

TypeScript, Next.js, React

[Live Demo](#) • [GitHub](#)

- **Problem:** An AI assistant that remembers everything unconstrained risks two failures: cross-person leakage (using A's information for B), and referencing memories that have expired or been revoked.
- **Role & Design:** Designed and built an Agent Memory layer prototype from zero to one. MemoryGate isolates memories by person and task scope; each memory carries source evidence, a confirmation status, a privacy tag, and a TTL – the model only ever sees the gate-filtered subset, never deciding for itself which memory to use.
- **Result:** Live, publicly accessible prototype; 16 synthetic scenarios cover cross-person leakage, sensitive information, and memory-expiry boundaries (deterministic prototype, not connected to a real chat platform or production LLM).

Technical Skills

Full-Stack Development: Python, TypeScript, Svelte, FastAPI, Node.js, React / Next.js, SQL, Git / GitLab, Linux

AI Applications: LLM application development, Agent Tool Use / Memory, retrieval & evidence citation, RAG selection judgment, evaluation design, Claude Code

Quality & Data: CI, Presto / Trino, Hive, PySpark, pandas, Google Drive / Sheets APIs

Languages: Native Chinese; English (IELTS 6.5), used for cross-country BI alignment in Singapore

Honors & Certifications: Software Design Engineer & Higher-Ed Teacher Qualification (China); Huawei ICT Competition Global Final, Cloud Track, Third Prize (top 5%, Jun 2022)